Introduction to Linux OS

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Comparison of Operating Systems

List of Linux Distributions

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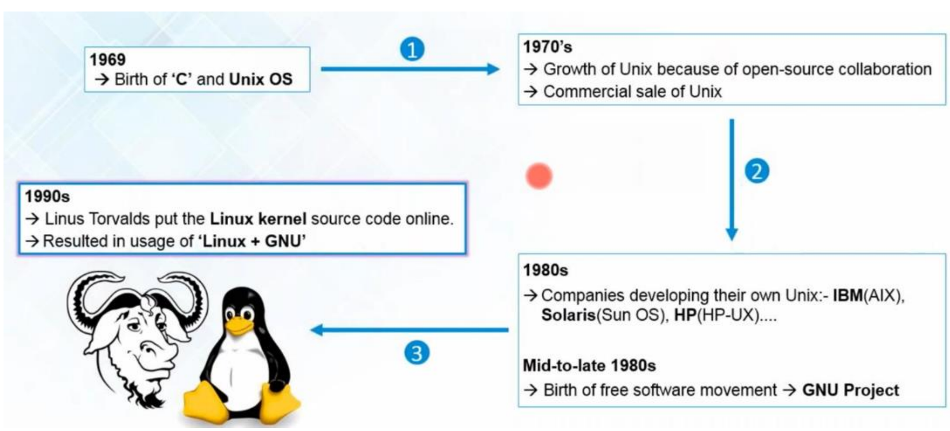
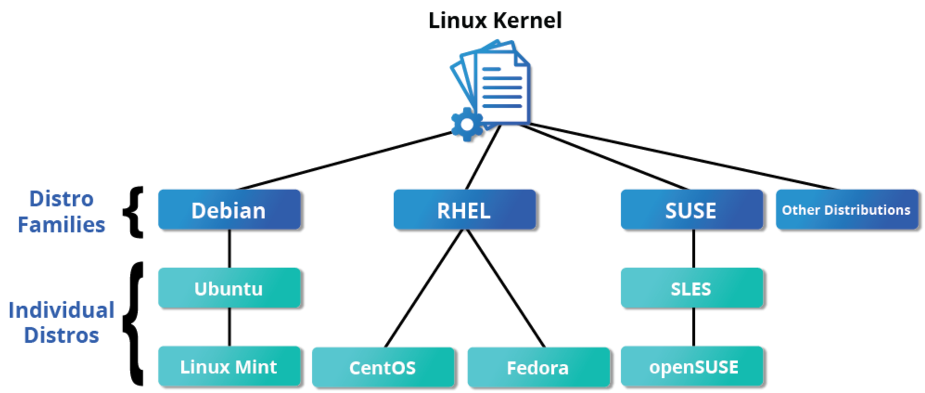
Linux History (What, Who, When, Where & Why Linux)

Linux is an open source computer operating system, initially developed on and for Intel x86-based personal computers. It has been subsequently ported to an astoundingly long list of other hardware platforms, from tiny embedded appliances to the world's largest supercomputers.

Linus Torvalds was a student in Helsinki, Finland, in 1991, when he started a project: writing his own operating system kernel. He also collected together and/or developed the other essential ingredients required to construct an entire operating system with his kernel at the center using C programming language. It wasn't long before this became known as the Linux kernel.

In 1992, Linux was re-licensed using the General Public License (GPL) by GNU (a project of the Free Software Foundation or FSF, which promotes freely available software), which enabled it to build a worldwide community of developers. By combining the kernel with other system components from the GNU project, numerous other developers created complete systems called Linux Distributions, which first appeared in the mid-90s.

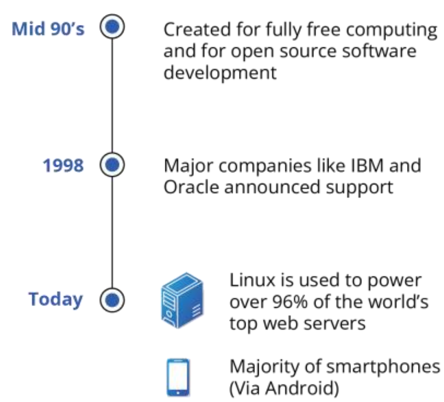




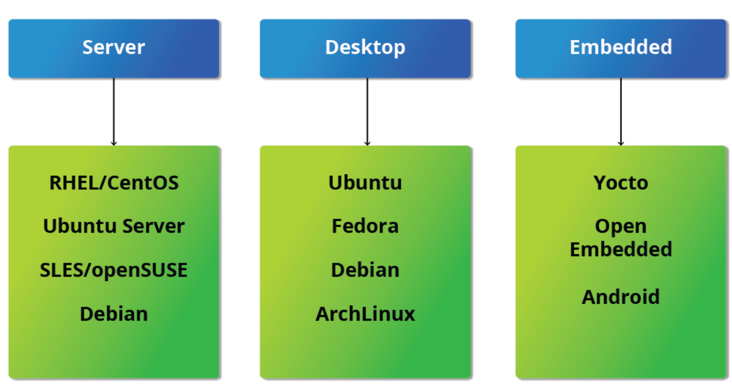
RHEL: Red-Hat Enterprise Linux

SUSE: Software und System-Entwicklung (i.e., Software and System-Design)

SUSE: Linux Enterprise Server



Choosing a Linux Distribution



What will we use? We'll use CentOS Stream 9 which is a RHEL based distribution.

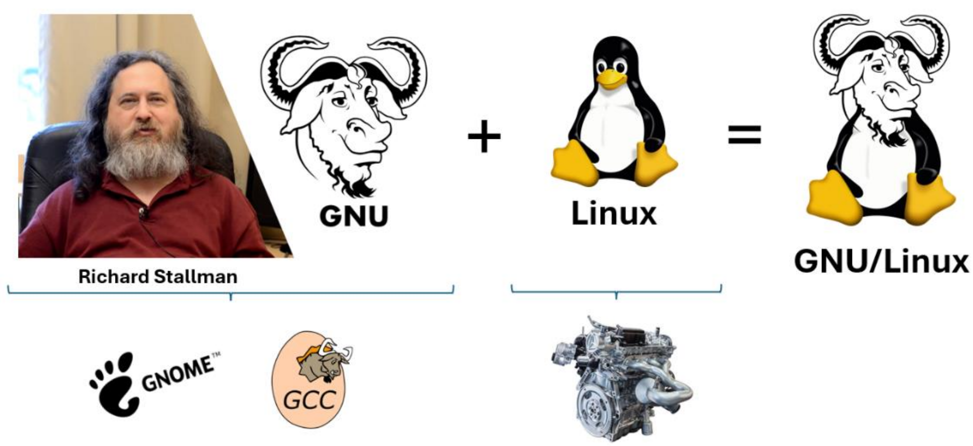
GNOME Desktop Environment

- [GNOME (GNU Network Object Model Environment)]() is a popular desktop environment with an easy-to-use graphical user interface used on various Unix-like operating systems, such as Linux distributions.
- It is bundled as the default desktop environment for most Linux distributions, including Red Hat Enterprise Linux (RHEL), Fedora, CentOS, SUSE Linux Enterprise, Ubuntu, and Debian.
- GNOME has menu-based navigation and is sometimes an easy transition to accomplish for Windows users.
- However, the look and feel can be quite different across distributions, even if they are all using GNOME.

Another common desktop environment very important in the history of Linux and also widely used is KDE (K Desktop Environment), which has often been used in conjunction with SUSE and openSUSE.

Other alternatives for a desktop environment include Unity (present on older Ubuntu but still based on GNOME), XFCE, and LXDE.

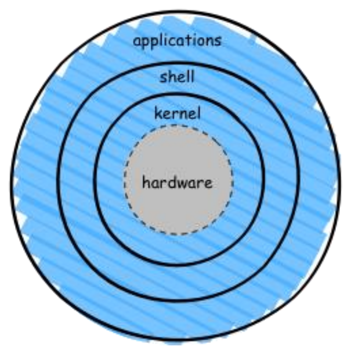
GNOME™



Introduction to the Command Line

Linux system administrators spend a significant amount of their time at the command line prompt. They often automate and troubleshoot tasks in this text environment. There is a saying, "graphical user interfaces make easy tasks easier, while command line interfaces make difficult tasks possible". Linux relies heavily on the abundance of command line tools. The command line interface provides the following advantages:

- No GUI overhead is incurred.
- Virtually any and every task can be accomplished while sitting at the command line.
- You can implement scripts for often-used (or easy-to-forget) tasks and series of procedures.
- You can sign into remote machines anywhere on the Internet.
- You can initiate graphical applications directly from the command line instead of hunting through menus.
- While graphical tools may vary among Linux distributions, the command line interface does not.



The Command Line

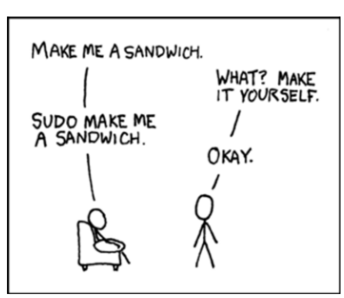
Most input lines entered at the shell prompt have three basic elements:

```
$ <command> <options> <arguments>
```

The command is the name of the program or script you are executing. It may be followed by one or more options (or switches) that modify what the command may do. Options usually start with one or two dashes, for example, -p or --print, in order to differentiate them from arguments, which represent what the command operates on. However, plenty of commands have no options, no arguments, or neither.

Sudo (SuperUser Do)

All the demonstrations created have a user configured with sudo capabilities to provide the user with administrative (admin) privileges when required. sudo allows users to run programs using the security privileges of another user, generally root (superuser).



Command	Usage
`su`	Login as a super user with high privileges (use 'exit' or 'CTRL + D' to exit sudo mode).
`man`	Displays the manual page for a command, providing detailed information on its usage. `$ man diff`
`init 6` `reboot`	Reboots (restarts) the system.
`init 0` `shutdown`	Shuts down the system.
`who`	Shows who is currently logged into the system.
`whoami`	Displays the username of the current user.
`last`	Displays a list of last logged in users.
`uname`	Prints system information.
`hostname`	Display or set the hostname of the system (`$ cat /etc/hostname`).
`hostnamectl`	A more comprehensive way to manage the system hostname and related metadata.
`history`	Shows the command history of the current session.
`passwd`	Change the current or other user's password.
`reset`	Resets the terminal.
`clear`	Clears the terminal screen.

`echo`	Prints a string in the terminal screen.
	`$ echo Hello`
	`$ echo -e "Ahmed\nMohamed\nMona"`

Hints:

- The `` `tab` `` button autocompletes commands and filenames/directories.
- Dealing with Long Shell Commands: the backslash (\ ) is the line continuation character.

```
$ echo "one \
two"
```